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(54) **EYE THERMAL COMPRESS**

(71) Applicant: **Umay Care Holdings Inc.**, Edmonton (CA)

(72) Inventor: **Ali Habib**, Edmonton (CA)

(73) Assignee: **Umay Care Holdings Inc.**, Edmonton (CA)

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(56) **References Cited**

U.S. PATENT DOCUMENTS

3,768,485 A * 10/1973 Linick A61F 7/103 607/109

6,155,995 A 12/2000 Lin

(Continued)

FOREIGN PATENT DOCUMENTS

CN 1297719 A 6/2001

CN 103800128 A 5/2014

(Continued)

OTHER PUBLICATIONS

European Communication in counterpart European Patent Application No. 17798433.3 issued on Jan. 25, 2024 (8 pages, in English).

(Continued)

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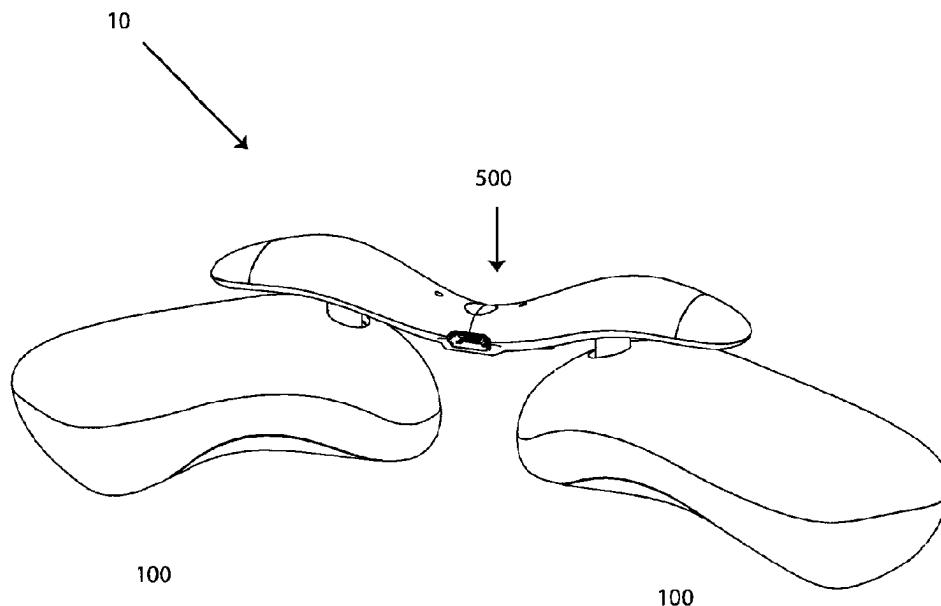
Assistant Examiner — Bradford C. Blaise

(74) *Attorney, Agent, or Firm* — Bookoof McAndrews, PLLC

(57) **ABSTRACT**

An eye compress assembly includes two thermally adjustable heating and cooling device and a bridge connecting the two devices is provided. The thermally adjustable device is configured to apply thermal treatment against the eye region. Within the thermally adjustable heating and cooling device the thermoelectric element is powered by a battery. A power input port is provided in the bridge to connect into a power supply in order to recharge the battery.

8 Claims, 10 Drawing Sheets



Related U.S. Application Data		2015/0190607 A1 *	7/2015	Sugio	A61B 5/6803 600/27
(60)	Provisional application No. 62/321,976, filed on Apr. 13, 2016.	2016/0000600 A1 *	1/2016	Lee	A61F 7/00 607/109
		2016/0106576 A1 *	4/2016	Badawi	A61F 7/02 607/109
(52)	U.S. Cl.	2017/0014300 A1	1/2017	Dippo et al.	
	CPC A61F 2007/0004 (2013.01); A61F 2007/0075 (2013.01); A61F 2007/0086 (2013.01); A61F 2007/0228 (2013.01); A61F 2007/0261 (2013.01)	2018/0256432 A1	9/2018	Mayo et al.	
		2021/0128058 A1 *	5/2021	Habib	A61B 5/4884
FOREIGN PATENT DOCUMENTS					
(58)	Field of Classification Search	CN	104207878 A	12/2014	
	CPC A61F 2007/0228; A61F 2007/0261; A61F 7/007; A61F 7/02; A61F 9/04	CN	104473759 A	4/2015	
	See application file for complete search history.	CN	104510558 A	4/2015	
		CN	205698210 U	11/2016	
		CN	106214330 A	12/2016	
		DE	202013102909 U1	9/2013	
(56)	References Cited	JP	2006198249 A	8/2006	
	U.S. PATENT DOCUMENTS	WO	2015030560 A1	3/2015	
		WO	2018044877 A1	3/2018	
	6,908,195 B2 *	6/2005	Fuller	A61H 35/02 351/158	
	11,696,850 B2 *	7/2023	Habib	A61F 7/02 607/108	
	2003/0195438 A1	10/2003	Petillo		
	2006/0200052 A1 *	9/2006	Lin	A61H 23/0263 601/37	
	2007/0282405 A1	12/2007	Wong et al.		
	2012/0222192 A1	9/2012	Carey et al.		
	2013/0172829 A1	7/2013	Badawi et al.		
	2014/0277303 A1	9/2014	Biser et al.		
	2015/0057701 A1 *	2/2015	Kelleher	A61H 23/0245 606/204.15	
OTHER PUBLICATIONS					
PCT International Search Report and Written Opinion, Application No. PCT/CA2017/050464, dated Aug. 1, 2017.					
Supplementary European Search Report, Application No. 17798433.3, dated Feb. 6, 2020.					
Extended European Search Report, Application No. 17798433.3, dated Jun. 24, 2020.					
Machine translation of the document "First Office Action for Chinese Application No. 20178002325", dated Jul. 1, 2020; pp. 1-9.					
* cited by examiner					

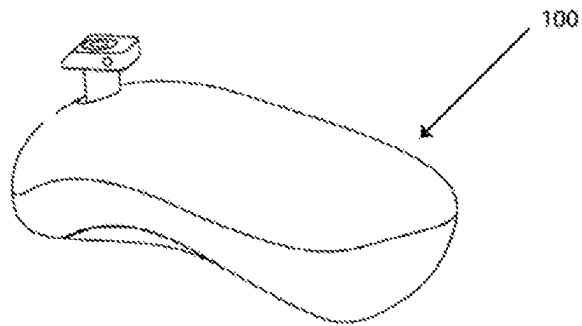


Figure 1

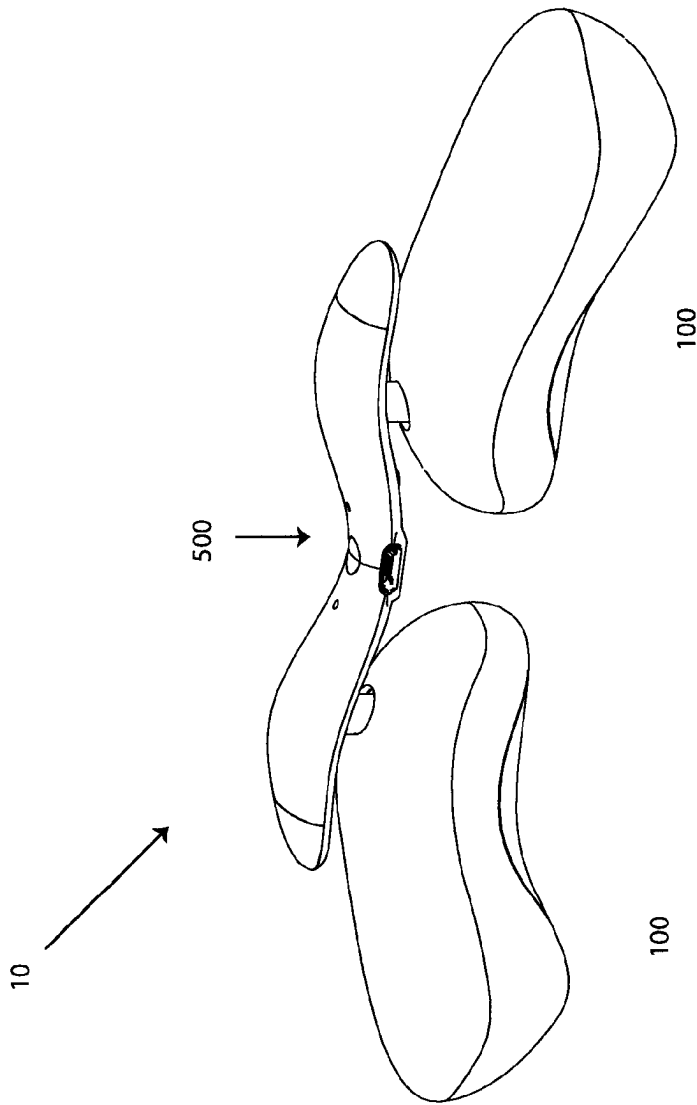


Figure 2

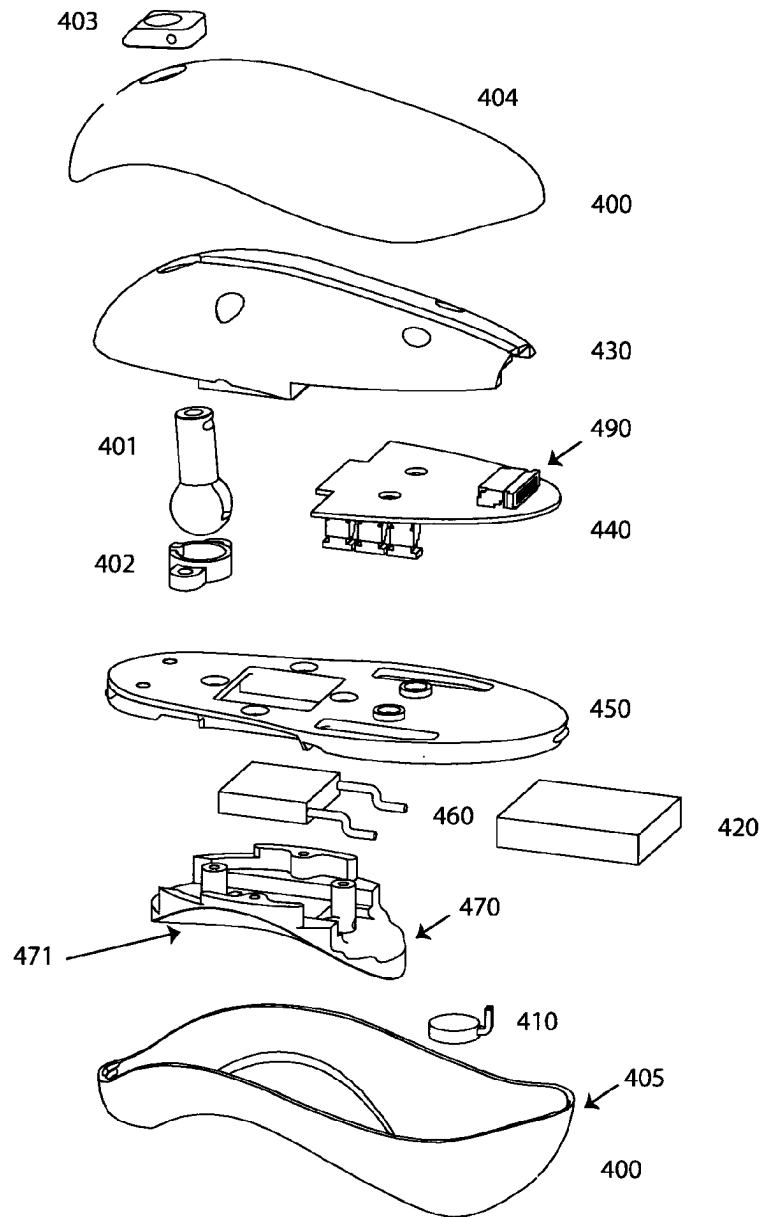


Figure 3

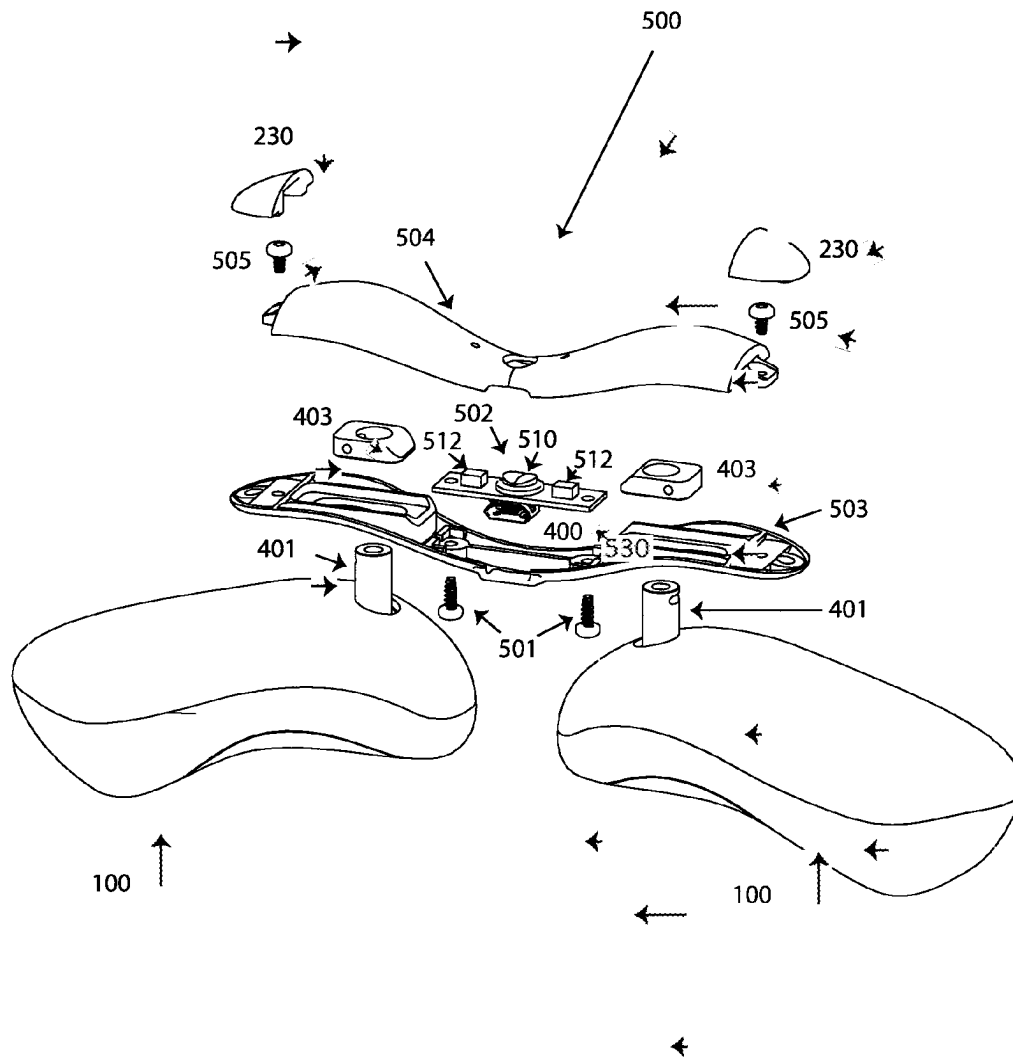


Figure 4

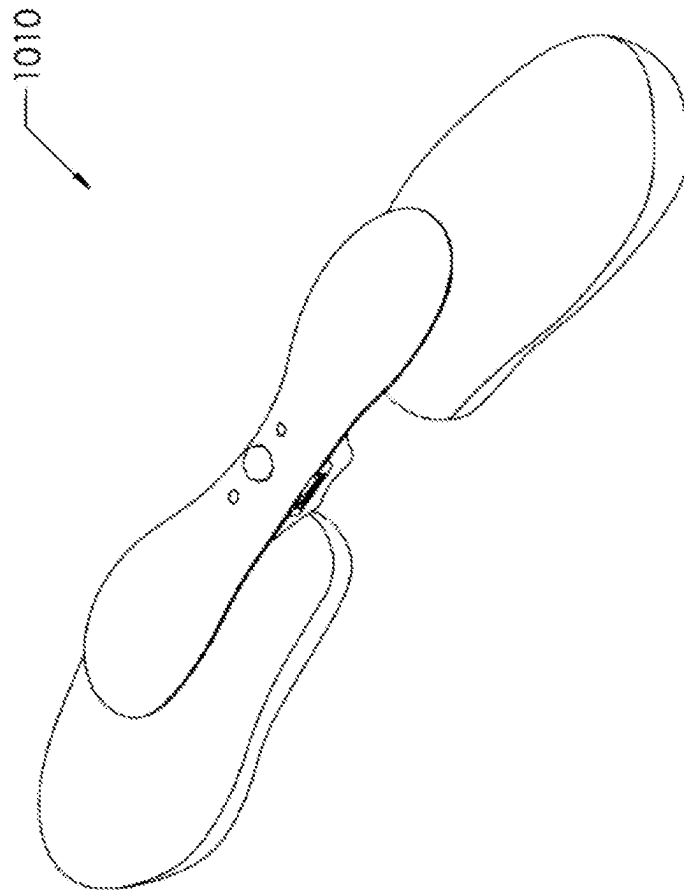
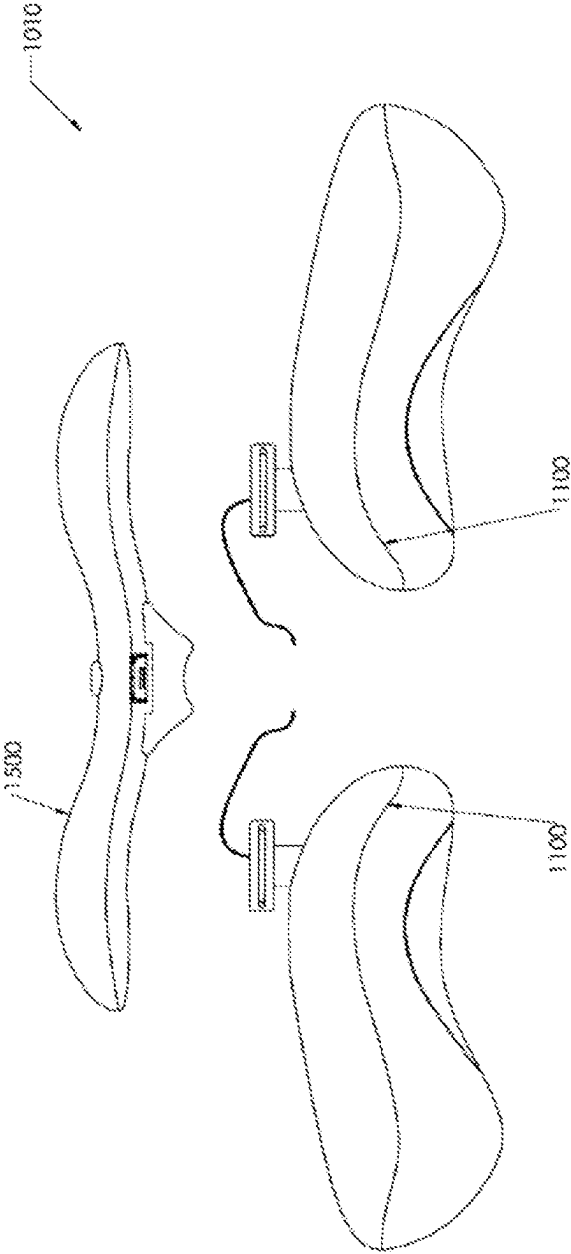


FIG 5



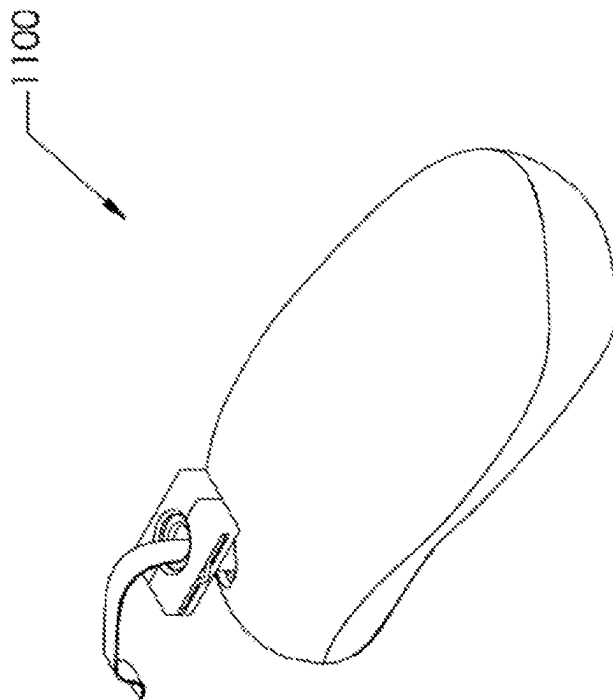
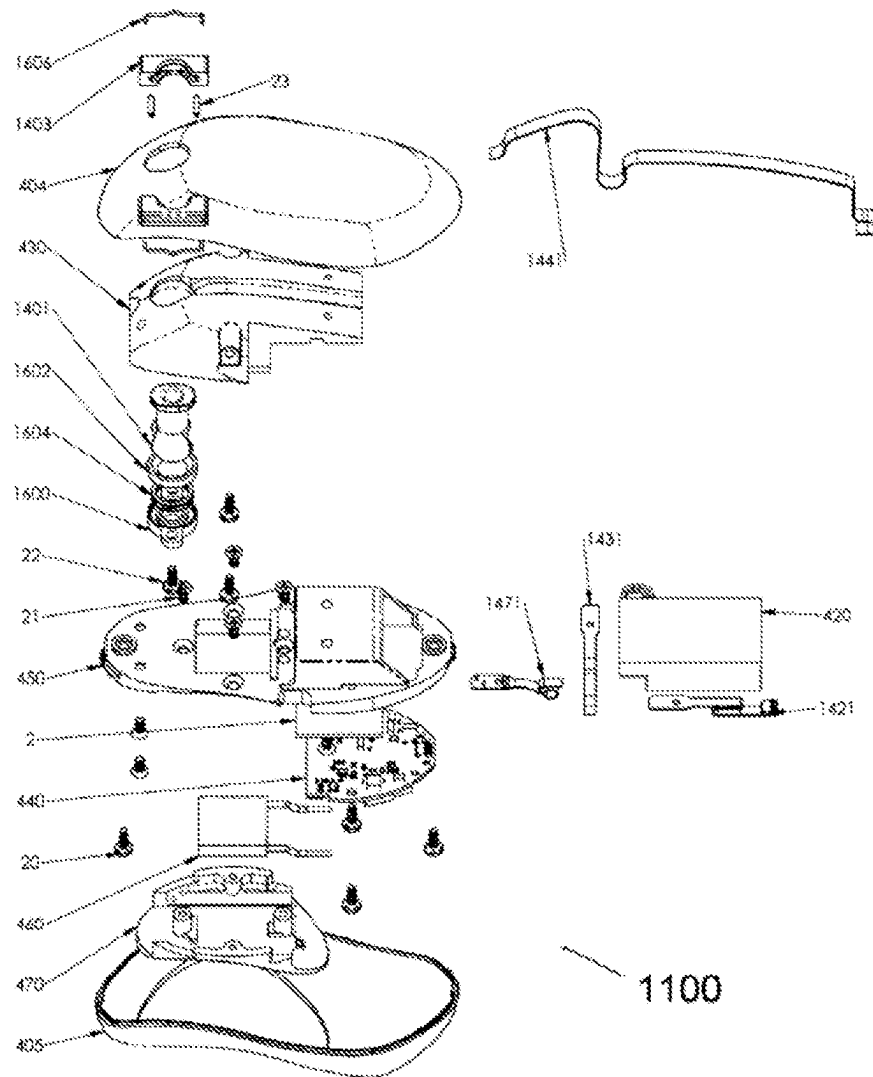


FIG 7



PG 2

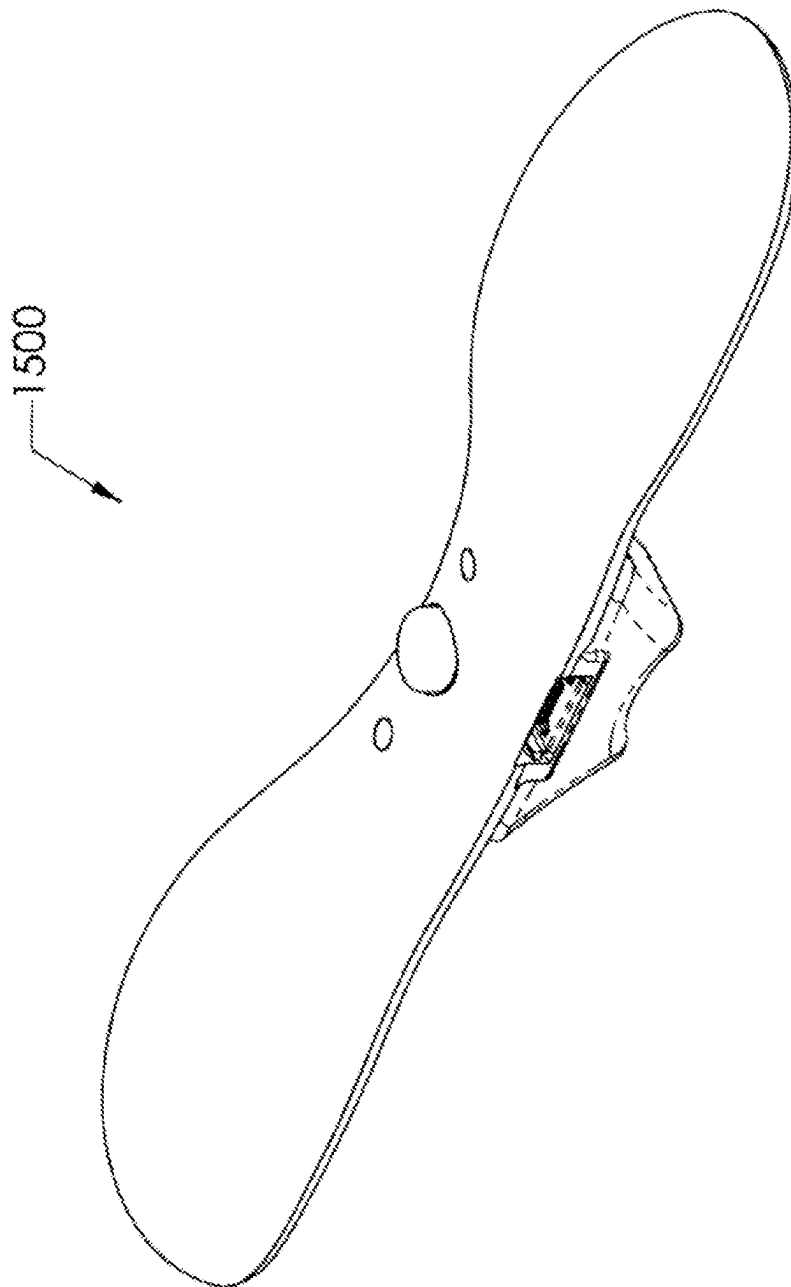


FIG 9

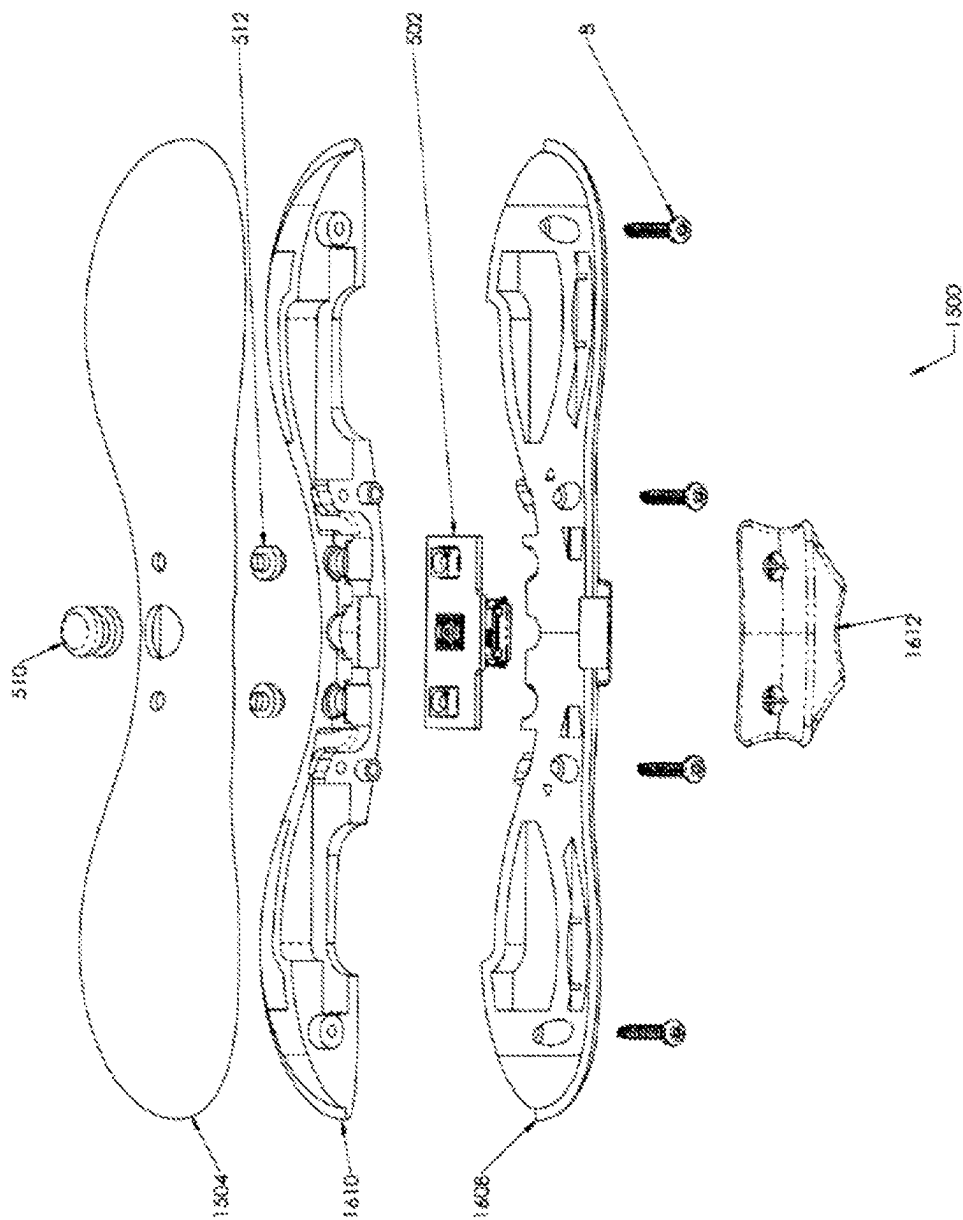


FIG. 10

EYE THERMAL COMPRESS**FIELD OF THE INVENTION**

The invention is directed to thermal health devices, and more specifically to thermal health devices for application to eyes.

BACKGROUND OF THE INVENTION

In the human eye, the tear film covering the ocular surfaces is composed of three layers. The innermost layer in contact with the ocular surface is the mucus layer. The mucus layer is comprised of many mucins. The middle layer comprising the bulk of the tear film is the aqueous layer. The aqueous layer is important in that it provides a protective layer and lubrication to prevent dryness of the eye.

Dryness of the eye can cause symptoms such as itchiness, burning, and irritation, which can result in discomfort. The outermost layer is comprised of many lipids known as "meibum" or "sebum." This outermost lipid layer is very thin, typically less than 250 nm in thickness. The lipid layer provides a protective coating over the aqueous and mucus layers to limit the rate at which these underlying layers evaporate. A higher rate of evaporation of the aqueous layer can cause dryness of the eye. Thus, if the lipid layer is not sufficient to limit the rate of evaporation of the aqueous layer, dryness of the eye may result. The lipid layer also lubricates the eyelid during blinking, which prevents dry eye. If the lipid layer can be improved, the rate of evaporation is decreased, lubrication is improved, and partial or complete relief of the dry eye state is achieved.

Environmental conditions can contribute to dry eye, for example, exposure to smoke, wind and dry climates can increase tear evaporation resulting in dry eye symptoms. Failure to blink regularly, such as when staring at a computer screen for long periods of time, can also contribute to drying of the eyes.

Dry eye can also be caused by a condition known as meibomian gland dysfunction (MGD). Symptoms of MGD include dryness, burning, itching, stickiness, watering, light sensitivity, red eyes, foreign body sensation, chalazions, styes, and intermittent blurry vision. Known treatments for MGD generally apply significant heat in order to melt, loosen, or soften obstructions and/or occlusions in the meibomian glands.

The application of hot and/or cold compresses is a known therapeutic treatment for some physical ailments. Some conventional methods of thermal compress therapy include a user holding a cloth (e.g., a washcloth) under hot or cold running water and then applying the moist, temperature-adjusted cloth to the desired body part. In some such instances, the cloth is maintained in contact with the desired body part through manual intervention (e.g., the user holds the cloth in place).

In other conventional methods, a silica gel (e.g., sodium acetate) pack can be heated or cooled, to apply thermal energy to or remove thermal energy from the eye region of the user. In some embodiments, the thermal packs can be included in a device that includes a container or frame configured to support the thermal packs and a strap system configured to retain the thermal packs in a fixed position relative to the eye region of the user. The anatomy of the eye region, however, can result in challenges to the application of thermal packs. For example, the contour of the eye region can result in challenges to placing the thermal packs with a relatively consistent and comfortable amount of force. As

such, the level of discomfort and/or ineffective application or removal of thermal energy can, in some instances, deter a user from using some such devices.

Regarding electrical heaters, one conventional eye treatment is described in U.S. Patent Pub. No. 2007/0060988. The heater it describes applies heat by using an electrical signal requiring the use of a thermocouple and sophisticated feedback control system to monitor and adjust the electrical signal to maintain heat between 43 and 47° C. to one eye for between 1 and 10 minutes. Furthermore, the device uses a screw to adjust pressure on the eye. Since it requires 1) a threaded shaft or screw adjustment, 2) elevated heat, and 3) precise thermal regulation independent of temperature, the time of treatment, actual temperature, and pressure on the eye must be administered and monitored by a medical physician or technician to avoid burning the eyelid or damaging the eye itself.

Another conventional heater is described in U.S. Pat. No. 4,261,364. The heater it describes uses a battery operated surgical heater that warms a compress resembling an eye patch for post-ophthalmic surgery patients. The heater is strapped to a surgical compress that applies heat to a patient's eye socket. Since the heater 1) is in molded plastic not integrated with the compress, 2) is battery operated, 3) uses wiring for a heating element, and 4) heats a compress rather than an eyelid, the result is an uncomfortable, uncontrolled heat source that cannot carefully control the temperature that reaches the eyelid itself. Due to these factors, the time of treatment, actual temperature, and pressure on the eye must also be administered and monitored by a medical physician or technician to avoid burning the eyelid or damaging the eye itself.

Thus, improved apparatus, systems and methods for providing thermal therapeutics to the eye region are needed.

SUMMARY OF THE INVENTION

The present application relates generally to apparatuses, systems and methods for applying thermal treatment to the eye region by the application or removal of thermal energy.

According to an embodiment of the present application, there is provided an eye thermal compress assembly comprising: two thermal treatment units each having an upper shell and a lower shell, the two thermal treatment units generating thermal energy to provide a heated or cooled lower surface; a bridge connecting the two thermal treatment units. In some embodiments, the eye thermal compress assembly is used with the user lying face up and the eye thermal compress assembly resting on the user's eyelids. In some embodiments, an optional attachment means is provided for attaching the eye thermal compress assembly to the head, allowing use of the eye thermal compress assembly when the user is not lying down.

As well, according to an embodiment of the present application there is provided a thermal treatment unit, the thermal treatment unit comprising: an upper shell; a mounting board; a lower shell; a thermoelectric cooler (TEC) for generating thermal energy; a printed circuit board (PCB) for controlling the TEC; a heatsink for dissipating thermal energy into the upper shell.

According to an embodiment of the present disclosure, there is provided an eye thermal compress assembly comprising: two thermal treatment units, the two thermal treatment units generating thermal energy to provide a heated or cooled treatment surface for the eyelids of a user when worn; and a bridge connecting the two thermal treatment units.

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According to an embodiment of the present disclosure, there is provided a thermal treatment unit, comprising: an upper shell; a lower shell; a mounting board; a thermoelectric cooler (TEC) for generating thermal energy; a controller for controlling the TEC; a heatsink for dissipating thermal energy into the upper shell; wherein the TEC, the heatsink and the controller are contained with inside the upper shell and the lower shell.

BRIEF DESCRIPTION OF THE DRAWINGS

Embodiments of the present invention will now be described by way of example with reference to attached figures, wherein:

FIG. 1 illustrates a side view of one assembled thermal treatment unit;

FIG. 2 illustrates a side view of an eye thermal compress assembly comprising two thermal treatment units and a bridge;

FIG. 3 illustrates an exploded view the thermal treatment unit according to an embodiment of the present application;

FIG. 4 illustrates a side exploded view of the bridge;

FIG. 5 is a perspective view of an eye thermal compress assembly according to another embodiment showing two thermal treatment units connected by a bridge;

FIG. 6 is a side elevation view of the eye thermal compress assembly of FIG. 5 showing the two thermal treatment units detached from the bridge;

FIG. 7 is a perspective view of the thermal treatment unit in isolation according to an embodiment;

FIG. 8 is an exploded view the thermal treatment unit of FIG. 7;

FIG. 9 is a perspective view of the bridge according to an embodiment; and

FIG. 10 is an exploded view the bridge of FIG. 9.

DETAILED DESCRIPTION OF EMBODIMENTS

Reference will be made below in detail to exemplary embodiments of the invention, examples of which are illustrated in the accompanying drawings. Wherever possible, the same reference numerals used throughout the drawings refer to the same or like parts.

FIG. 1 illustrates a side view of an assembled thermal treatment unit 100.

FIG. 2 illustrates a side view of the eye thermal compress assembly 10 according to an embodiment. The eye thermal compress assembly 10 includes two thermal treatment unit 100 connected by a bridge 500, so that the eye thermal compress assembly may apply heat therapy or cold therapy to both eyes at the same time if desired. In some embodiments, the user can choose to use either or both of the thermal treatment units at a time. The two thermal treatment units are identical to or as mirror image to each other. In one embodiment, the eye thermal compress assembly 10 is used while the user is lying face up and the thermal treatment unit 100 rests on the eyelids of the user without using any means to attach the eye thermal compress assembly 10 to the head of the user. This prevents too much pressure on the eyelids. It may also improve safety because it allows the eye thermal compress assembly to slide off the user's face when the user falls asleep. In some embodiments, an optional attachment means is used to secure the eye thermal compress assembly 10 in place in operation. The attachment means may be made from a flexible material such as neoprene that allows the attachment means to stretch. With the optional attachment means, the eye thermal compress assembly 10 does not

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require the user to lie down when using the eye thermal compress assembly 10. In some embodiments, there is provided an optional flexible band attached to the ends of the attachment means for attaching the eye thermal compress assembly 10 around a human head in operation more securely. The eye thermal compress assembly 10 does not require a medical physician or technician for administration and is adjustable for different wearers.

An optional moistened pad can be used between user's closed eyelid(s) and the thermal treatment units 100. Any suitable moistened pad known in the field may be used. In some embodiment, the moistened pad may be made of a fibrous non-woven fabric or other soft cloth material. In some embodiments, the moistened pad 220 is sized for a typical adult. In some embodiments, ointment may be provided on the moistened pad. In some embodiments, one or more pharmacological agent is provided in the pad. In some embodiments, the moistened pad is removably attached to the lower surface of the thermal treatment unit 100.

FIG. 3 illustrates an exploded view of the one of the unassembled thermal treatment unit 100. The thermal treatment unit 100 includes an upper shell 404, a heatsink 430, a lower shell 405 which forms a lower surface and which contacts a portion of a user's face, a mounting board 450, a haptic motor 410, a battery 420, a component 470, thermoelectric cooler (TEC) 460, a first PCB 440, a first receptacle 402, a connector 401 and a second receptacle 403.

When in operation, the lower shell 405 is close to the face of the user while the upper shell 404 is away from the face of the user. The heatsink 430 is close to the upper shell 404. In some embodiments, the heatsink 430 is disposed next to the upper shell 404 and the upper part of the heatsink 430 is shaped to fit the interior of the upper shell 404.

A pin connection component 490 may be used to connect the TEC 460 to the first PCB 440 by electrical wire (not shown). In some embodiments, the battery 420 is connected to the first PCB 440. The first PCB 440 may be used to control the battery 420 and TEC 460 during use. In some embodiments, the PCB 440 may control charging of the battery 420. The first PCB 440 may be connected to the battery 420 and TEC 460 by any suitable means known in the field, for example, by using electrical wires (not shown). The battery 420 may be rechargeable and/or replaceable. In a preferred embodiment, the battery is rechargeable and fixed. The battery 420 provides power to the TEC 460.

In some embodiments, the first PCB 440 may be connected to the haptic motor 410. The haptic motor 410 may be used to provide notifications to the user by way of vibration. The haptic motor 410 may be placed anywhere within the treatment unit so long as it can provide the notification. In one embodiment, the haptic motor 410 is attached to the side of the lower shell 405 that is interior to the thermal treatment unit 100 by adhesive. The haptic motor 410 may be attached by any other means known in the field. For example, a structure may be configured within the thermal treatment unit 100 to receive the haptic motor 410.

The mounting board 450 is used to secure the different components within the thermal treatment unit 100, for example, the PCB 440. In some embodiments, a hole is configured in the mounting board 450 so that the upper part of the TEC 460 passes through the hole to be in contact with the lower part of the heatsink 430. The heatsink 430 is used to draw heat away from the TEC 460 during cooling treatment. In some embodiments, the lower part of the heatsink 430 may be formed to maximize the contact area with the TEC 460. In some embodiments, the battery 420 is attached to the bottom of the mounting board 450 by any

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suitable means known in the field. In some embodiments, the battery **420** is attached to the mounting board by adhesive.

The mounting board **450** is also used to fixedly receive a first receptacle **402**. The first receptacle **402** is for receiving the lower end of the connector **401**. In a preferred embodiment, the surface of the first receptacle **402** for receiving the lower end of the connector **401** is partially spherical, while the lower end of the connector **401** is ball-shaped, so that the lower end of the connector **401** forms a ball joint structure with the first receptacle **402**. In some embodiments, the upper part of the connector **401** is a stem that extends through holes in the heatsink **430** and the upper shell **404**. In some embodiments, the connector **401** is hollow to allow electrical wires therethrough to connect the first PCB **440** to the bridge **500**.

In some embodiments, the component **470** has a treatment surface **471** that is in contact with the eyelid when the thermal treatment unit **100** is in operation. In some embodiments, the treatment surface **471** is eye shaped and sized for a typical adult. In a preferred embodiment, the treatment surface **471** is smooth. The smooth treatment surface **471** may be made of metal, plastic, rubber, textile or any other suitable material. Preferably, the smooth treatment surface **471** is made of metal, selected from steel, stainless steel, copper, silver, gold and aluminum. In some embodiments, the lower shell **405** has a hole configured to expose the surface **471** to the exterior of the lower shell **405** when the thermal treatment **100** is assembled, such that the surface **471** is in contact with the eyelid when the thermal treatment unit **100** is in operation.

In some other embodiments, the lower shell **405** is configured with an eye shaped portion sized for a typical adult on the lower exterior surface of the lower shell **405**. The eye shaped portion is in contact with the closed eyelids when the treatment unit **100** is in operation to provide heating or cooling treatment to the eyes. In such embodiments, the surface of the lower shell **405** that is interior of the lower exterior surface is in contact with the lower surface of component **470** to receive heating or cooling from the TEC **460**.

The component **470** may also be configured to provide a structure to receive the TEC **460** therein so that the TEC **460** is close to the smooth treatment surface **471** and that the TEC **460** will not move freely inside the thermal treatment unit **100**. In some embodiments, the structure provided is a groove to fit the lower part of the TEC **460** therein. In some embodiments, when the thermal treatment unit **100** is assembled, the component **470** is fixedly attached to the mounting board **450**. The attachment can be by any suitable means known in the field. In preferred embodiments, the attachment is by adhesive and/or screws.

In operation, the thermal treatment units **100**, which are battery powered, are placed over a closed outer eyelid to provide controlled heat or cold therapy. The thermal elements **100** may be made of a smooth material that is safe for placement against the skin. The thermal treatment unit **100** may include redundant temperature sensors (not shown) to ensure precise temperature control. As well, the first PCB **440** may identify a minimum and maximum temperature that may be generated by the thermal treatment unit **100** to ensure that the thermal elements are within a safe temperature range for use against unprotected skin. The thermal treatment unit **100** transfers thermal energy (e.g. hot or cold) supplied from a thermoelectric chip, such as the TEC **460** and directs the thermal energy via component **470** towards the user's eyes and eyelids. The TEC **460** generates thermal

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energy within the thermal treatment unit **100**. The TEC **460** provides heating or cooling by means well-known in the field.

FIG. 4 illustrates a side exploded view of bridge **500**. The Bridge **500** connects the two thermal treatment units **100**. The bridge **500** comprises an upper shell **504** and a lower shell **503**. The upper shell **504** and the lower shell **503** can be connected by any suitable means known in the field. For example, FIG. 4 shows that they are connected by screws **505** and the screws **505** are then covered with screw covers **230**. The two thermal treatment units **100** are also electrically connected to the bridge **500** so that the bridge **500** can be powered by at least one battery **420** in one of the thermal treatment units **100**. The electrical connection can be by the electrical wires through the hollow connectors **401** as described earlier.

There are holes configured in the lower shell **503** so that the upper part of the connector **401** can extend through the holes. The upper end of **401** is then received within the second receptacle **403** so that the thermal treatment unit **100** is connected to the bridge **500**. The two thermal treatment units **100** are fixedly or adjustably attached to the bridge **500**. For example, if there is ball-joint structure between the connector **401** and the first receptacle **402**, the connection would be adjustable in relation to the thermal treatment unit **100**. In some embodiments, each of the holes in the lower shell **503** to receive each of the stems of the connectors **401** is a groove to allow movement of the upper part of the connector **401** so that the thermal treatment units **100** are adjustable in relation to the bridge, for example, to fit different sizes of faces of the users.

As well, the bridge **500** includes a power input port **530** for receiving an electrical cord to connect the bridge **500** with an electrical power supply, thus providing power for charging the batteries **420** of the thermal treatment units **100** and/or for operation of the bridge **500**. When an electrical cord is not connected to the power input port **530** of the bridge **500**, at least one of the batteries **420** of the two thermal treatment units **100** that are attached to the bridge **500** may provide power for the operation of the bridge **500**. In some embodiments, the power input port **530** is a micro USB port.

The bridge may further comprise a controller **502** so that the power input port **530** can be fixed to the controller **502**. Controller **502** comprises a processor, a memory for storing programs for instructing the processor, and input and output. The bridge **500** may further include a push button **510**, which may also be fixed to the controller **502**. The bridge **500** may further comprise at least one indicator light **512**, which may also be fixed to the controller **502**. In some embodiments, the button **510** and the light **512** are integrated. In some other embodiments, the button **510** and the light **512** are separate.

When the upper shell **504** and lower shell **503** are assembled, the controller **502** is fixedly sandwiched between the upper shell **504** and the lower shell **503**. The upper shell **504** and lower shell **503** are so configured that when they are assembled the power input port **530** is exposed for receiving a power input cord. The upper shell **504** may have holes configured therein for exposing the switch **510** and the indicator light **512**.

In some embodiments, the button **510** and indicator light **512** are electrically connected to the first PCB **440**. Control signals and power can be sent to the first PCB **440** and battery **420**, while power can be sent from either or both batteries **420** from the two thermal treatment unit **100** to the button **510** and the light **512**. The connection can be by any

suitable means in the field, for example, by electrical wires (not shown). In operation, the bridge **500** is a powered hardware interface that allows a user to control treatment cycles of the thermal treatment units **100**. For example, there may be a single 15-minute heat therapy treatment or a 5-minute cold therapy.

In a preferred embodiment, the controller **502** is a second PCB. In some embodiments, the button **510** and the light **512** may be fixed to the second PCB **502**. In such embodiments, the second PCB **502** may be electrically connected to the first PCB **440**, for example, by wire through the hollow **401** connector. The power input port **530** may be fixed to the second PCB **502**. In such embodiments, the power from the power input port **530** is sent through the second PCB **502** to the first PCB **440** for charging the battery **420** and control signal is also sent from the second PCB **502** to the first PCB **440**. When no external power is provided at the power input port **530**, the second PCB **502** may receive power from at least one of the batteries **420** from the two thermal treatment units.

The button **510** may be a switch or selector and can be used for selectively activating the heating or cooling of the thermal treatment units, for controlling charging the battery within the treatment units, and for deactivating the heating or cooling of the treatment units. For example, depression of the push button **510** may send signal to the first PCB **440** of at least one of thermal treatment units **100** to activate the preheating (or precooling) sequence or programs of the thermal treatment units **100**. The activation of the sequence may be indicated with a one-motor vibration generated by the motor **410** in the thermal treatment unit **100**. This may be followed by a two-motor vibration that indicates that therapeutic temperature of the thermal treatment unit **100** has been reached and a three motor vibration when the heat or cooling therapy has completed. The push button **510** can be depressed again at any time to stop the treatment, which immediately stops the therapeutic heat (or cooling) application. As well, the indicator light **512** may be used for indicating activation of the thermal treatment, the status of the thermal treatment and charging of the thermal treatment units by the lights **512** changing color and/or flashing in different patterns. In some embodiments, the second PCB **502** is configured to store pre-determined heating and/or cooling sequences that the user can choose from by pressing the button **510**. In embodiments in which the button **510** is fixed to the second PCB **502**, the operation of the button **510** and light **512** described above may be realized through the second PCB **502** by the second PCB **502** providing suitable control mechanisms.

FIGS. **5** to **10** show another embodiment of the eye thermal compress assembly **1010** according to the present disclosure. The eye thermal compress assembly **1010** includes two thermal treatment units **1100** connected by a bridge **1500**. Each thermal treatment unit **1100** may include upper shell **404**, heatsink **430**, lower shell **405**, mounting board **450**, battery **420**, component **470**, thermoelectric cooler (TEC) **460**, and first PCB **440**. Pin connection component **490** may be used to connect TEC **460** to first PCB **440** by an electrical wire **1441**.

Each thermal treatment unit **1100** may further comprise a first receptacle **1402**, a ball joint **1401**, and a second receptacle **1403**. Ball joint **1401** includes a first and second opposed ends and defines a wire channel dimensioned to allow passage of electrical wire **1441** therethrough.

First receptacle **1402** is configured to receive first end of ball joint **1401** and comprises a ball joint retainer **1600**, a ball joint friction plate **1602** and a wave disk spring **1604**.

Wave disk spring **1604** may be received in a channel formed in ball joint retainer **1600**. Ball joint friction plate **1602** and wave disk spring **1604** cooperate to provide for increased resistance of the first end of ball joint **1401** when it is inserted into first receptacle **1402**.

Second receptacle **1403** is configured to receive second end of ball joint **1401**. Second receptacle **1403** may be rectangular with chamfered edges or may be trapezoid with beveled edges. A spring **1606** may be provided to increase resistance of the second end of the ball joint **1401** when it is inserted into second receptacle **1403**.

Each thermal treatment unit **1100** may further comprise additional components for appropriate temperature control such a heatsink thermistor **1431**, a component thermistor **1471**, and a battery thermistor **1421**.

As shown in FIGS. **9** and **10**, a bridge **1500** connects the two thermal treatment units **1100**. Bridge **1500** comprises an upper shell **1504**, a lower shell **1503**, and PCB **502** enclosed by upper **1503** and lower shell **1503**. Lower shell **1503** may comprise a first frame **1608** and a second frame **1610**. Frame **1608** and **1610** are configured to matingly combined with each other to form lower shell **1503**.

Lower shell **1503** may be configured provide a support structure for PCB **502**. Formed in lower shell **1503** are grooves **1612** dimensioned to receive second receptacle **1403** so as to permit sliding movement of second receptacle **1403** therein. Formed in at least one of frame **1608** or **1610** is a power input port **1530** for receiving an electrical cord (not shown) to connect PCB **502** of bridge **1500** with a source of electrical power for charging battery **420**.

Bridge **1500** may further comprise a nose piece **1612** configured for placement onto the nose when assembly **1010** is in use. As shown, nose piece **1612** may be removeably secured to lower shell **1503**. For additional comfort, the positioning of nosepiece **1612** may be adjusted to accommodate different shaped and sized noses as necessary.

Once a preheating (or precooling) sequence has been activated and the two thermal treatment units **100** or **1100** have reached the desired temperature, the two thermal treatment units **100** or **1100** are placed over the closed eyelids. Alternatively, the eye thermal compress assembly can first be put on the eyelids of the user lying face up and then gradually heating or cooling to the desired temperature.

Variations may be made to some example embodiments, which may include combinations and sub-combinations of any of the above. The various embodiments presented above are merely examples and are in no way meant to limit the scope of this disclosure. Variations of the innovations described herein will be apparent to persons of ordinary skill in the art having the benefit of the example embodiments, such variations being within the intended scope of the present disclosure. In particular, features from one or more of the above-described embodiments may be selected to create alternative embodiments comprised of a sub-combination of features which may not be explicitly described above. In addition, features from one or more of the above-described embodiments may be selected and combined to create alternative embodiments comprised of a combination of features which may not be explicitly described above. Features suitable for such combinations and sub-combinations would be readily apparent to persons skilled in the art upon review of the present disclosure as a whole. The subject matter described herein intends to cover and embrace all suitable changes in technology.

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Certain adaptations and modifications of the described embodiments can be made. Therefore, the above discussed embodiments are considered to be illustrative and not restrictive.

The invention claimed is:

1. An adjustable eye thermal compress assembly comprising:

two separate, electrically controllable and individually adjustable thermal treatment units for generating thermal energy to provide a heated or cooled treatment surface for a user's eyelids when worn; and

a bridge connecting the two thermal treatment units, each thermal treatment unit including:

a connector configured for connecting with the bridge via a groove in the bridge such that the connector and the associated thermal treatment unit are configured to move laterally with respect to the bridge;

an upper unit shell; and

a lower unit shell that, when coupled with the upper unit shell, forms an interior space, wherein each of the thermal treatment units is laterally adjustable relative to the bridge and rotatably adjustable relative to the bridge.

2. The assembly of claim 1, wherein the bridge comprises an upper bridge shell and a lower bridge shell, and the lower bridge shell comprises at least two holes forming portions of the grooves, wherein two of the at least two holes are provided for connection to the respective connectors of the associated thermal treatment units.

3. The assembly of claim 1, wherein the interior space comprises: a mounting board; a thermoelectric cooler (TEC) for generating thermal energy; and a heatsink for dissipating thermal energy.

4. The assembly of claim 3, further comprising one or more of the following: a motor attached to an inner side of

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the lower unit shell for producing a vibration notification, a pin connection component, for electrically connecting the TEC to a controller within the bridge, a treatment surface at an exterior surface of the lower unit shell, and a battery within the interior space for powering the TEC.

5. The assembly of claim 3, further comprising a pin connection component, for electrically connecting the TEC to a controller.

6. The assembly of claim 1, further comprising a controller to store programs of heating and/or cooling, wherein the controller is provided in the bridge or in at least one of the thermal treatment units.

7. The assembly of claim 6, wherein a switch is provided for choosing the programs of heating and/or cooling.

8. An adjustable eye thermal compress assembly comprising:

two separate, electrically controllable and individually adjustable thermal treatment units for generating thermal energy to provide a heated or cooled treatment surface for a user's eyelids when worn; and

a bridge connecting the two thermal treatment units, each thermal treatment unit including:

a connector;

an upper unit shell; and

a lower unit shell, that when coupled with the upper unit shell, forms an interior space,

wherein each of the thermal treatment units is laterally adjustable relative to the bridge, each thermal treatment unit including a ball joint, a first receptacle and a second receptacle, a lower end of the connector including a ball of the ball joint received by the first receptacle, an upper end of the connector is a stem, and the upper end of the connector is received by the second receptacle for attachment to the bridge.

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